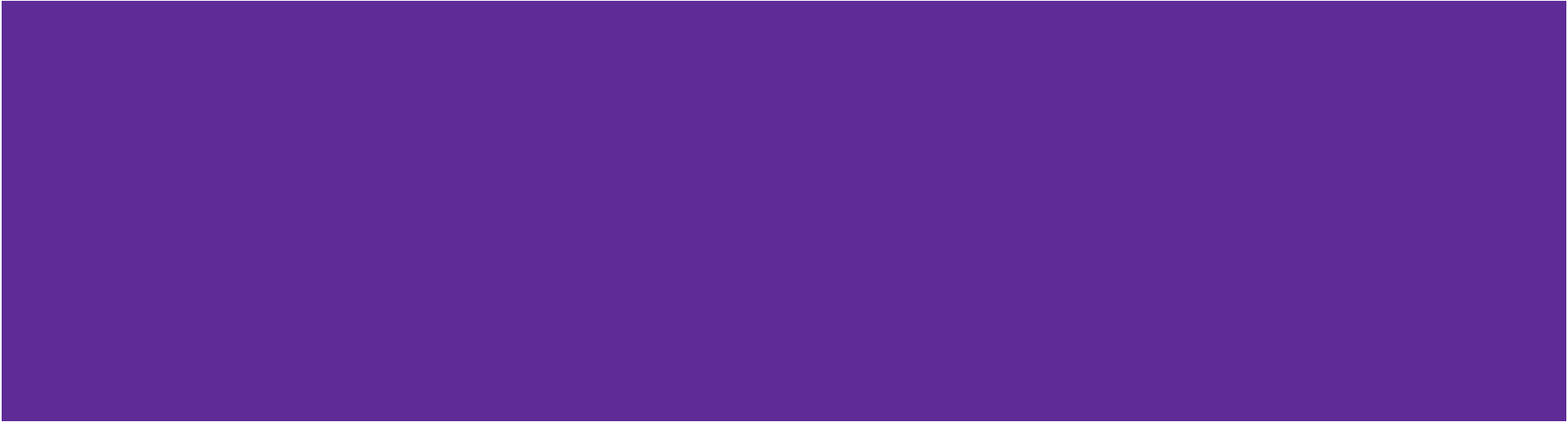


BrightLearn Data Datawarehouse Build

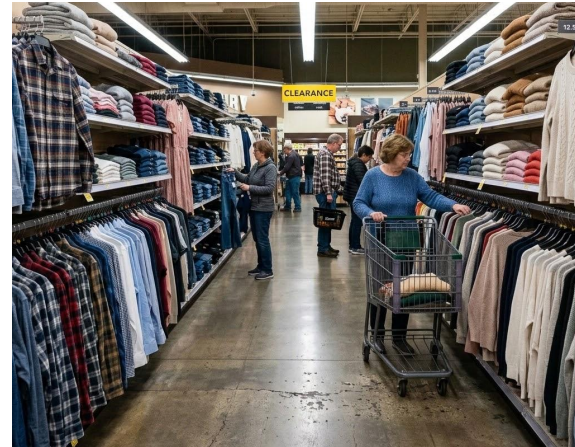
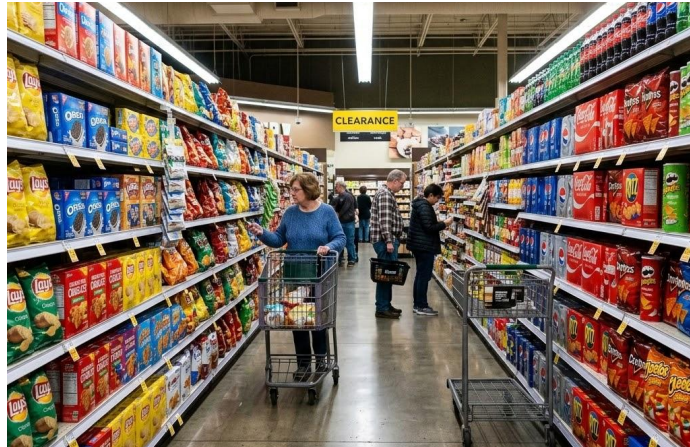
By Mpho Matebesi
Junior Data Engineer



Project Background

BrightLearn retail chain: Operates five stores across South Africa, selling Groceries, Electronics, Clothing, Home & Kitchen, and Beauty & Personal Care, with a three-tier loyalty programme (Bronze, Silver, Gold) rewarding repeat purchases.

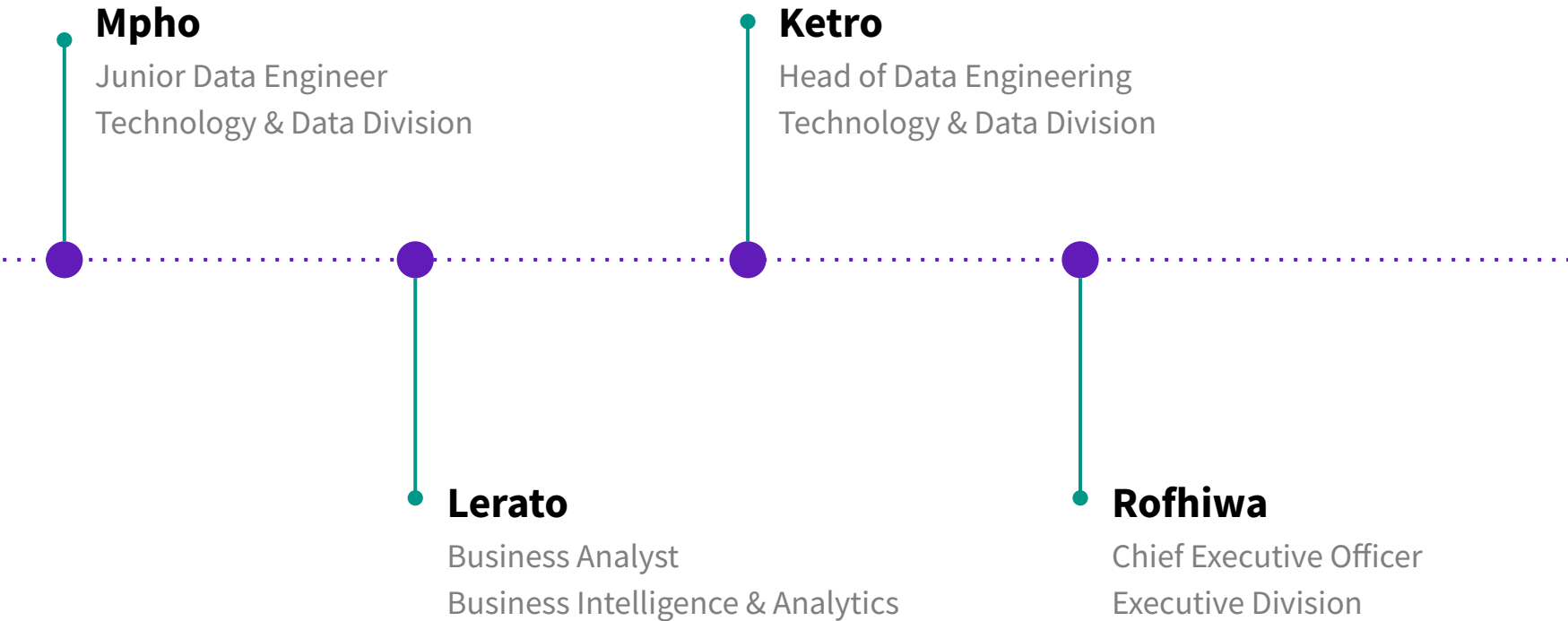
Store locations and systems: Stores are in Sandton, Cape Town, Umhlanga, Pretoria, and Gqeberha, each running its own POS system to record customer transactions.



Skills & expertise

- Star Schema
- SSMS
- Business Analytical Questions
- SSIS
- Presentation

BrightLearn Team



Mpho

Junior Data Engineer
Technology & Data Division

Ketro

Head of Data Engineering
Technology & Data Division

Lerato

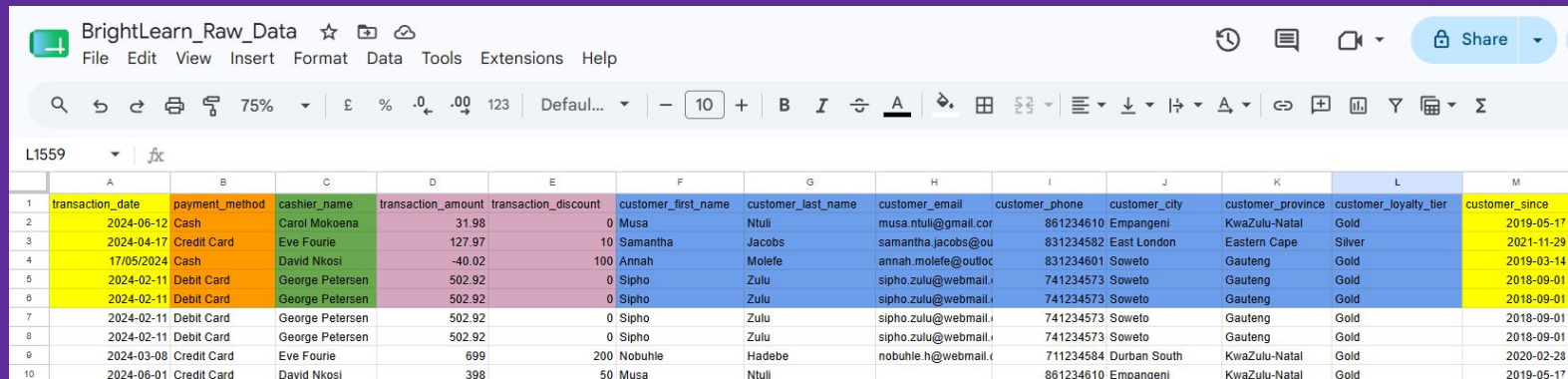
Business Analyst
Business Intelligence & Analytics

Rofhiwa

Chief Executive Officer
Executive Division

Receiving the CSV

- **Data Source:** BrightLearn_Raw_Data.csv .
- **Initial Observation:** Unstructured raw export containing transactional, customer, and store details.
- **Key Issues Spotted:** Missing Customer values, duplicate transactions, inconsistent product names/casing.
- **Action:** Ingested the raw CSV file into a staging database environment to begin data profiling and inspection.

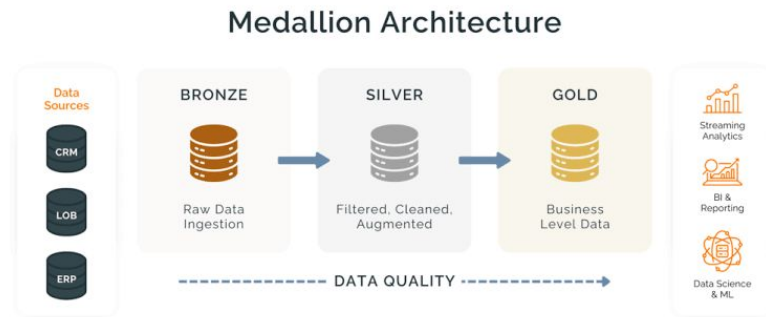


	A	B	C	D	E	F	G	H	I	J	K	L	M
1	transaction_date	payment_method	cashier_name	transaction_amount	transaction_discount	customer_first_name	customer_last_name	customer_email	customer_phone	customer_city	customer_province	customer_loyalty_tier	customer_since
2	2024-06-12	Cash	Carol Mokoena	31.98	0	Musa	Ntuli	musa.ntuli@gmail.com	861234610	Empangeni	KwaZulu-Natal	Gold	2019-05-17
3	2024-04-17	Credit Card	Eve Fourie	127.97	10	Samantha	Jacobs	samantha.jacobs@ou	831234582	East London	Eastern Cape	Silver	2021-11-29
4	17/05/2024	Cash	David Nkosi	-40.02	100	Annah	Molefe	annah.molefe@outloc	831234601	Soweto	Gauteng	Gold	2019-03-14
5	2024-02-11	Debit Card	George Petersen	502.92	0	Sipho	Zulu	sipho.zulu@webmail.i	741234573	Soweto	Gauteng	Gold	2018-09-01
6	2024-02-11	Debit Card	George Petersen	502.92	0	Sipho	Zulu	sipho.zulu@webmail.i	741234573	Soweto	Gauteng	Gold	2018-09-01
7	2024-02-11	Debit Card	George Petersen	502.92	0	Sipho	Zulu	sipho.zulu@webmail.i	741234573	Soweto	Gauteng	Gold	2018-09-01
8	2024-02-11	Debit Card	George Petersen	502.92	0	Sipho	Zulu	sipho.zulu@webmail.i	741234573	Soweto	Gauteng	Gold	2018-09-01
9	2024-03-08	Credit Card	Eve Fourie	699	200	Nobuhle	Hadebe	nobuhle.h@webmail.i	711234584	Durban South	KwaZulu-Natal	Gold	2020-02-28
10	2024-06-01	Credit Card	David Nkosi	398	50	Musa	Ntuli		861234610	Empangeni	KwaZulu-Natal	Gold	2019-05-17

Medallion Architecture Planning

Medallion Architecture Planning

- **Bronze Layer (Raw):** Direct ingestion of raw CSV data without modification; preserves the source record context.
- **Silver Layer (Cleansed/Staging):** Cleaned, filtered, and standardized staging tables with correct data types and mapped surrogate keys.
- **Gold Layer (Curated Warehouse):** Production-ready dimensional model featuring fully populated Fact and Dimension tables.
- **Value:** Ensures structured incremental refinement, fault tolerance, and clear lineage.

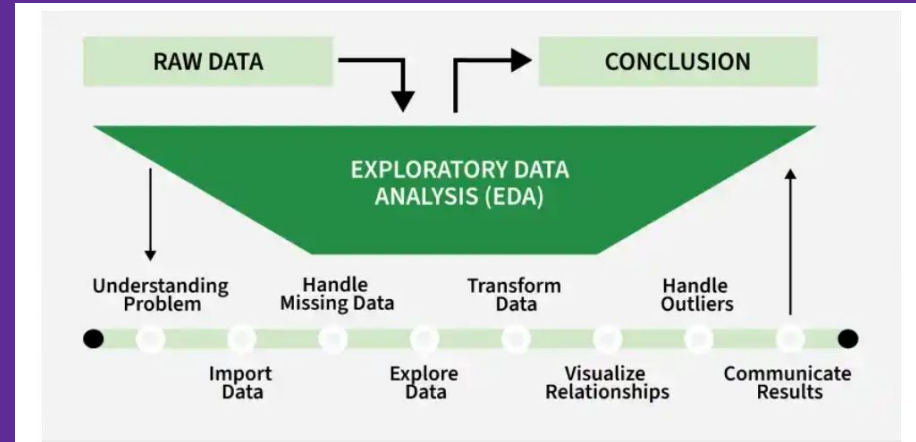


Loading CSV with SSIS

- **Integration Tool:** Built using SQL Server Integration Services (SSIS).
- **Ingestion Strategy:** Configured **Flat File Connection Manager** to read and parse the delimiter-separated input file.
- **Staging Area:** Import flat file rows into `stg_bright_mart_sales.dbo.bright_mart_raw_data` staging tables within SSMS.
- **Validation:** Verified initial column data type mappings (converting text strings to proper datatypes such as INT, DECIMAL, and DATETIME)

Exploratory Data Analysis (EDA)

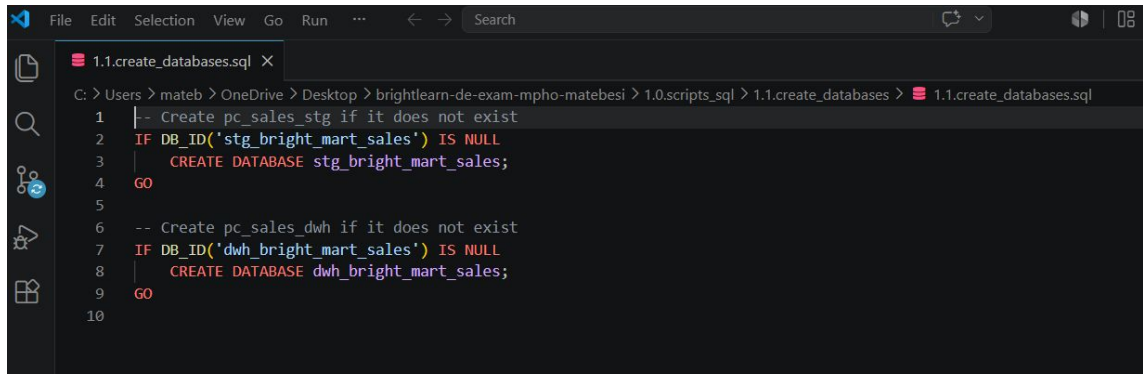
- **Profiling Findings:**
 - **124** missing CustomerID records (unauthenticated checkouts).
 - **87** duplicate transaction rows.
 - **42** negative or impossible stock-on-hand values.
- **Impact:** Identified data anomalies that would skew revenue metrics and inventory counts if left unaddressed.
- **Documentation:** Quantified and categorized all anomalies to feed into the formal Data Quality Report.



Data Modeling (Star Schema)

- **Fact Table (BrightMart's_fact_table):** Stores core numeric business metrics (Qty, Unit_price, Cost_price, Line_amount, Stock_on_hand, Reorder_threshold, Transaction_amount, Transaction_discount).
- **Dimension Tables:**
 - dim_store (Store details, location, manager, cashier).
 - dim_product (Supplier, product names, categories, SKU).
 - dim_customer (Customer contact info, loyalty tiers).
 - dim_date (Transaction date, customer history context).
 - dim_payment (Payment methods).
- **Benefit:** Optimized for analytical performance, easy joins, and fast business intelligence querying.



A screenshot of a code editor window showing a SQL script. The editor has a dark theme and a sidebar on the left with icons for Explorer, Search, Source Control, and Run and Debug. The file explorer shows the path: C:\Users\mateb>OneDrive>Desktop>brightlearn-de-exam-mpho-matebesi>1.0.scripts_sql>1.1.create_databases>1.1.create_databases.sql. The script content is as follows:

```
1  |-- Create pc_sales_stg if it does not exist
2  IF DB_ID('stg_bright_mart_sales') IS NULL
3  |   CREATE DATABASE stg_bright_mart_sales;
4  GO
5
6  -- Create pc_sales_dwh if it does not exist
7  IF DB_ID('dwh_bright_mart_sales') IS NULL
8  |   CREATE DATABASE dwh_bright_mart_sales;
9  GO
10
```

Creating the Databases

- **Database Name:** stg_bright_mart_sales & dwh_bright_mart_sales
- **Schema Design:** Structured under the dbo schema with explicit data typing.
- **Integrity Constraints:**
 - Assigned Primary Keys (PK) to all dimension tables.
 - Defined Foreign Keys (FK) within the fact table referencing dimension keys to enforce strict referential integrity.
 - Mind the database your working within

Staging Dimensions

- **Population:** Extracted distinct attributes from staging to populate dim_store, dim_product, dim_customer, dim_date, and dim_payment.
- **Standardization:**
 - Cleaned inconsistent strings (e.g., standardized variations into "Samsung Galaxy A14").
 - Consolidated regional and city naming conventions.
- **Lookup Logic:** Built lookup transformations in SSIS to resolve dimensional surrogate keys dynamically.



Cleaning Data into Warehouse

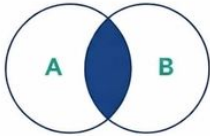
- **Deduplication:** Filtered out the identified duplicate transaction rows.
- **Flagging Negative Transactions:** 1 means True (Negative/Adjustment), 0 means False (Normal Sale)
- **Invalid Record Handling:** Redirected corrupted dates and missing critical keys.
- **Outcome:** Guaranteed that only high-quality, cleansed records entered the target Gold data warehouse.



SQL JOINS

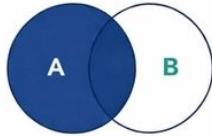
A = Students (Left Table) | B = Subjects (Right Table)

INNER JOIN



Returns only matching records in A and B
Only students who are also subjects

LEFT JOIN



Returns all records from A, and matching from B
All students, even if they don't match a subject

RIGHT JOIN



Returns all records from B, and matching from A
All subjects, even if no student is enrolled

Fact Table & Joins

- **Data Loading:** Populated Bright Mart's `_fact_table` with FK references to dimensions and validated line amounts.
- **Relational Joins:** Executed multi-table LEFT JOIN queries across `dim_store`, `dim_product`, and `dim_date`.
- **Operational Insight Example:** Identified stock-outs by joining stores and products where `Stock_on_hand <= Reorder_threshold`.

Solving Business Questions

- **Reorder Thresholds:** Pinpointed exact product-store pairs currently needing inventory replenishment.
- **Stock-Out Analysis:** Identified stores with the highest frequency of inventory shortages across regions.
- **Top Performers:** Aggregated total revenue ($\text{Line_amount} / \text{Transaction_amount}$) to rank top-selling products, categories, and payment methods.
- **Strategic Impact:** Directly informs purchasing, logistics, and localized marketing campaigns.



Building the SSIS Pipeline

- **Control Flow Architecture:** Truncate Staging, Extract Raw CSV, Transform & Cleanse Load Dimensions then Load Fact Table.
- **Data Flow:** Dynamic mapping from Flat File Source through Lookup & Derived Column transformations to OLE DB Destinations.
- **Fault Tolerance:** Built automated conditional split outputs to capture bad records into audit tables without breaking pipeline execution.

Building an ETL Pipeline Using SSIS



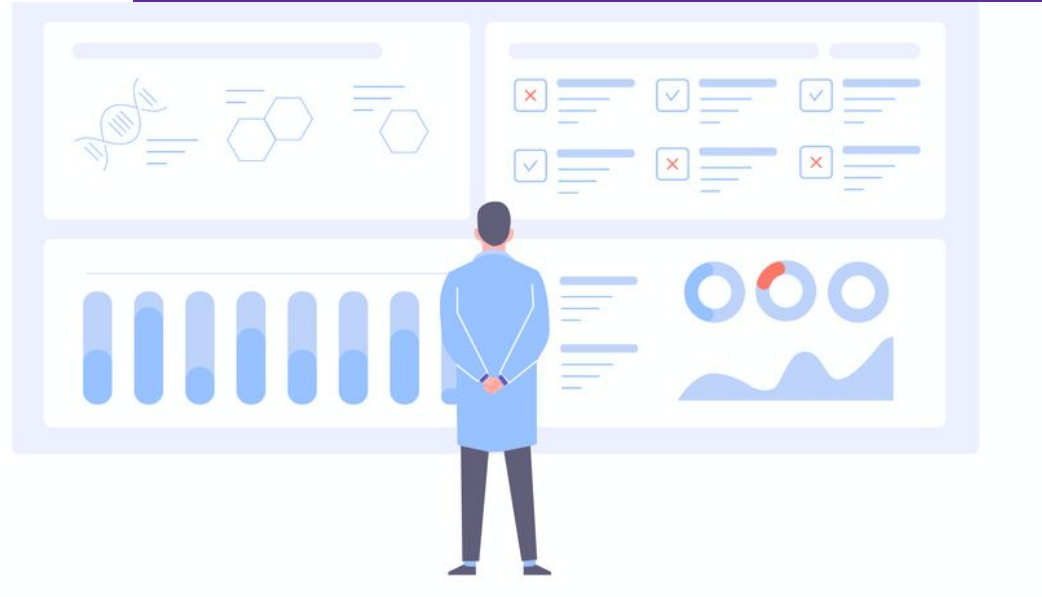
Data Quality Report

- **Documented Metrics:**
 - Full log of missing IDs, duplicated transaction IDs, invalid date strings, and stock outliers.
 - Quantified impact on total ingested volume.
- **Resolution Action Log:** Recorded all data treatment rules applied during the SSIS transformation phase.
- **Version Control:** Stored alongside code artifacts within the project's GitHub repository.



Final Deliverables

- **Data Warehouse:** Fully functional *dwh_bright_mart_sales* SQL Server database utilizing a clean Star Schema.
- **Automated Pipeline:** SSIS package configured for repeatable ETL processing.
- **Repository & Reporting:** Complete GitHub repo containing DDL/DML scripts, SSIS project files, Data Quality Report, and Power BI dashboards.



THE END

